

(a) Design A: 4 temperatures $\times$ 3 densities, $n = 1$

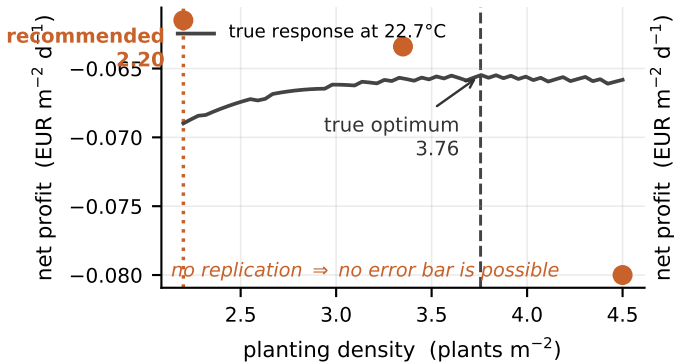
loop 0	18°C 2.20 -0.339	23°C 2.20 -0.061	27°C 2.20 -0.143	32°C 2.20 -0.151
loop 1	18°C 3.35 -0.335	23°C 3.35 -0.063	27°C 3.35 -0.154	32°C 3.35 -0.170
loop 2	18°C 4.50 -0.339	23°C 4.50 -0.080	27°C 4.50 -0.175	32°C 4.50 -0.184
	chamber 0	chamber 1	chamber 2	chamber 3

rejected: pure-error df = 0(b) Design B: split-plot, 2 temperatures $\times$ 3 densities, $n = 2$

loop 0	22°C 2.66 -0.092	28°C 2.66 -0.166	28°C 2.66 -0.166	22°C 3.35 -0.083
loop 1	22°C 4.04 -0.081	28°C 3.35 -0.181	28°C 4.04 -0.182	22°C 4.04 -0.100
loop 2	22°C 3.35 -0.081	28°C 4.04 -0.192	28°C 3.35 -0.181	22°C 2.66 -0.094
	chamber 0	chamber 1	chamber 2	chamber 3

accepted: pure-error df = 6

(c) Design A has no way to check itself



(d) Design B can separate chamber from treatment

